ModCRE: a structure homology-modeling approach to predict TF binding in cis-regulatory elements - https://sbi.upf.edu/modcre

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Introduction

Transcription factor (TF) binding is a key component of genomic regulation. Therefore, experimental methods exist for their identification. These, however, remain exacting and consequently their use is limited. For instance, the binding specificities of 25% human TFs remain uncharacterized; they do not have a known (i.e. determined experimentally) or computationally inferred motif. Here, we introduce ModCRE, a web server implementing a structure homology-modelling approach to predict TF binding motifs in order to identify cis-regulatory elements and automatically model higher-order TF regulatory complexes. ModCRE predicts a set of TF binding PWMs of a submitted TF structure or sequence and scans DNA for these. Then, matches are either profiled with a binding score or collected for their integration into a model containing DNA, TFs and transcription co-factors. In this paper we demonstrate how integrating protein-binding microarray (PBM) data addresses the protein-DNA structure scarcity issue in calculating statistical potentials, which in turn are shown to serve as capable predictors of TF binding PWMs. We reveal the conditional advantage of a structure-based over the nearest-neighbor approach in predicting TF binding sites as well as the improvement in both cases by means of rank-enrichment prediction. Finally, we apply ModCRE to automatically model the interferon beta enhanceosome, and the complex of the transcription factors SOX11 and SOX2 with the nucleosome.

Experimental PWM logos and best predictions

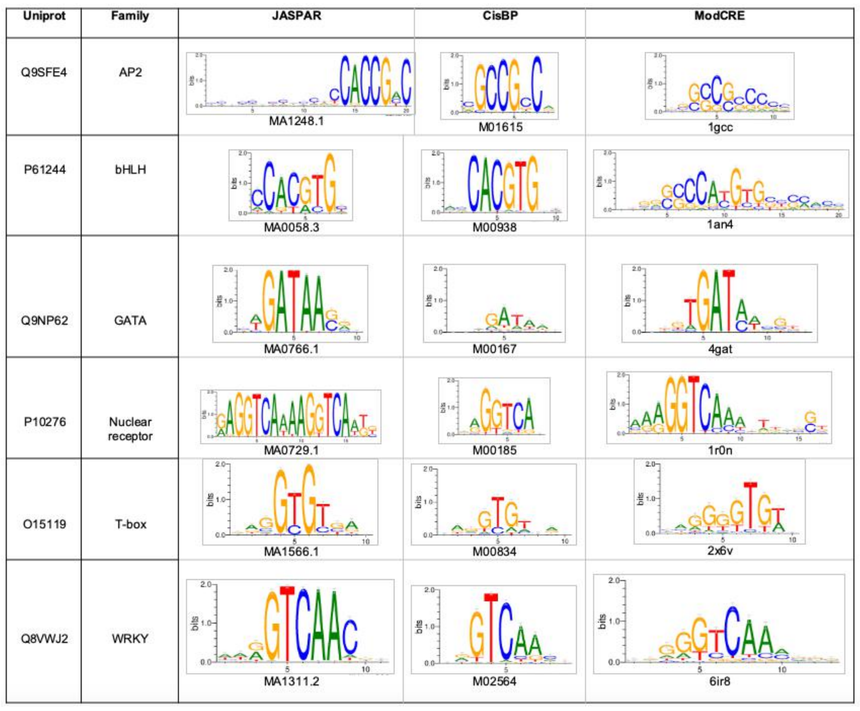

Figure 1: Predicted vs experimental PWMs

ModCRE vs the state-of-the-art approach

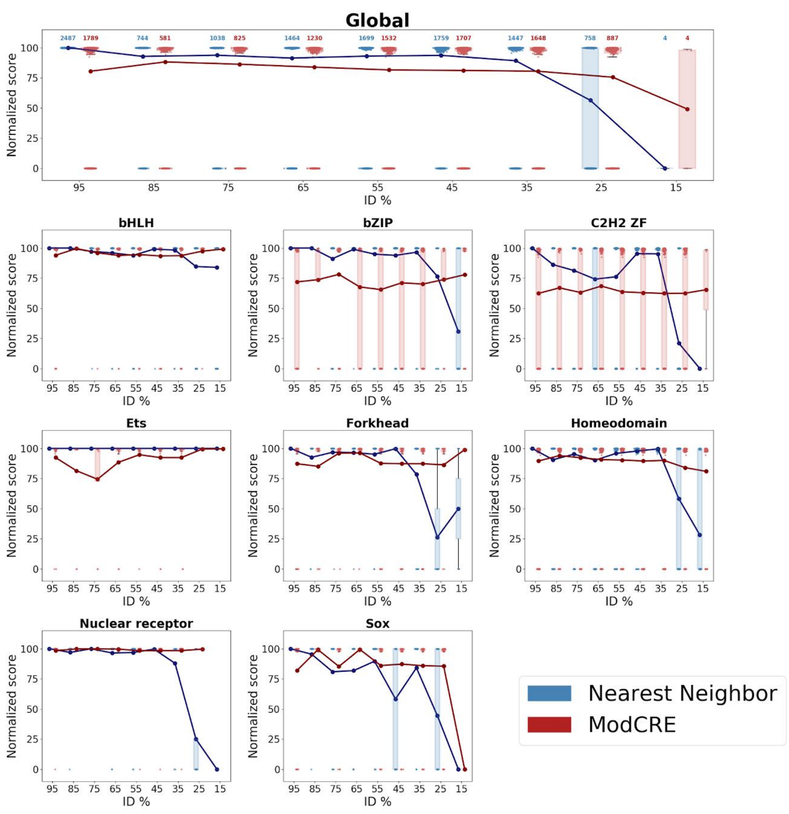

Figure 2: Distribution of the normalized ranking score of motif predictions with the nearest-neighbor approach and ModCRE using rank-enrichment. See (https://sbi.upf.edu/modcre/##faq) point 13.

Conclusions

- ModCRE outperforms the PWM prediction accuracy of the nearest neighbor approach at low levels of sequence identity in the most similar TF.
- The ModCRE approach can be applied to 88.7% of TF sequences.
- Due to advent of AlphaFold2[1,2] even unknown TF structures can be predicted and used in conjunction with ModCRE.

1. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. Nature 596, 583-589 (2021).
2. Jumper, J. & Hassabis, D. Protein structure predictions to atomic accuracy with AlphaFold. Nat Methods 19, 11-12 (2022).

Example ModCRE models

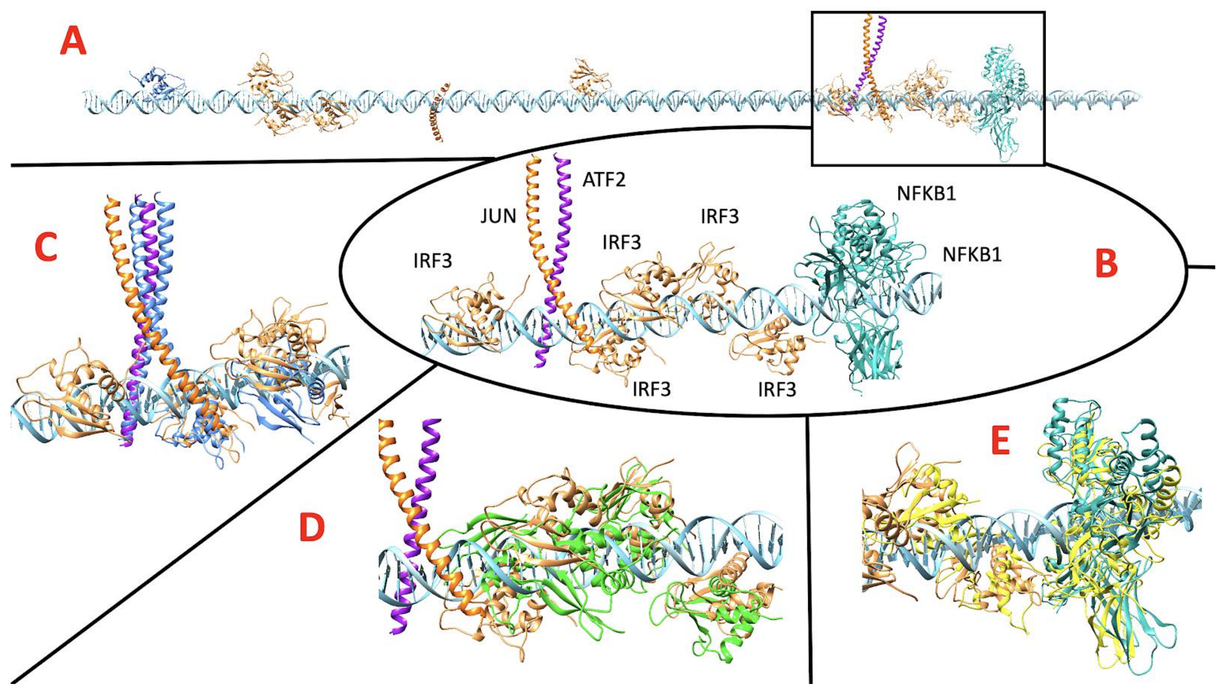

Figure 3: Modelled IFN-*beta* enhanceosome complex. A,B) predicted models. C,D,E) Superposition with Crystal structures (1T2K,2PI0,2O61 PDB)

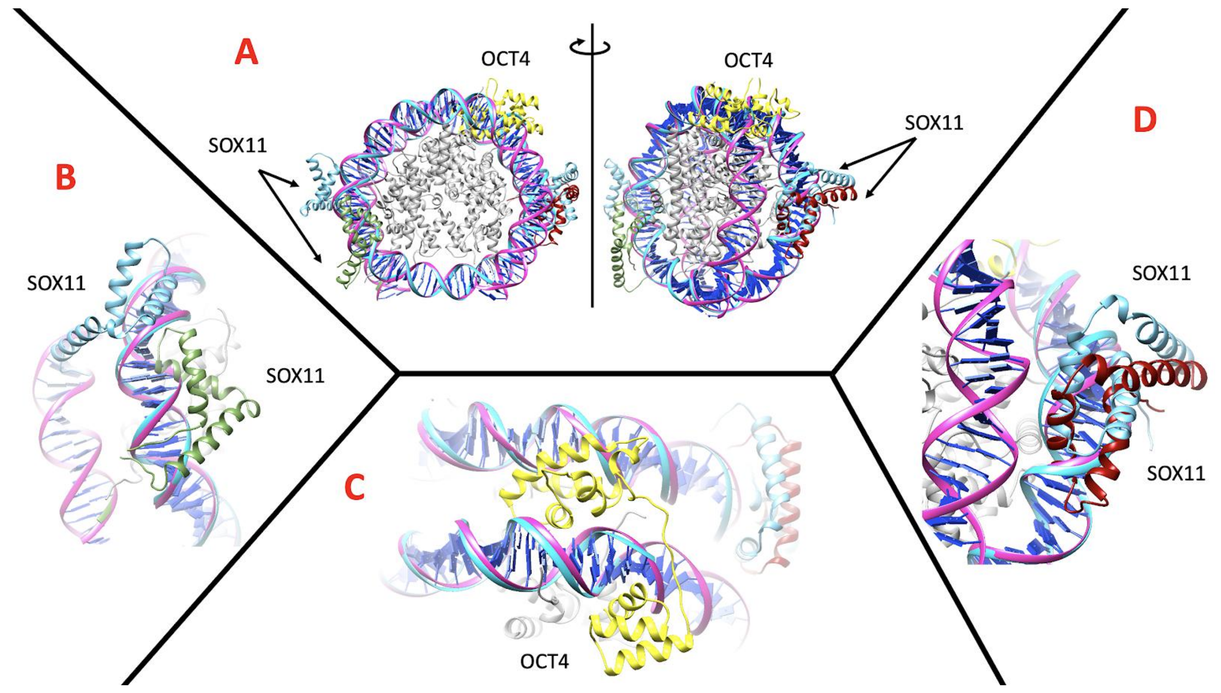

Figure 4: Modelled nucleosome complex with SOX11/SOX2 and OCT4 superposed with crystal structure (6T7C PDB). A) Front & lateral view. B) SOX11 54-64 bp segment. C) Oct4 binding. D) SOX11 86-96 bp segment.

ModCRE without available structure

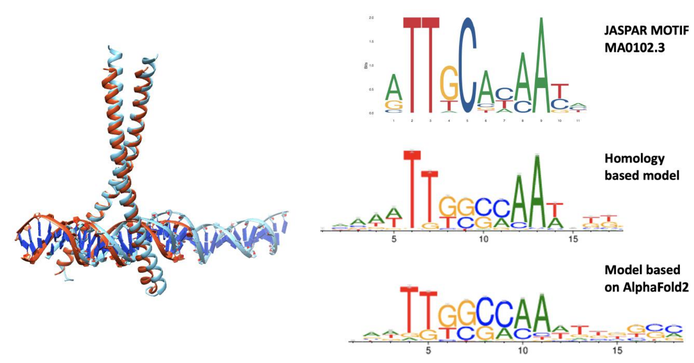

Figure 5: Prediction of PWMs of human C/EBPα and a superposition of a model of human C/EBPα (red) obtained with the crystal structure of rat (1NWQ PDB) as template and a ModCRE model based on the structure predicted with AlphaFold2 (blue)

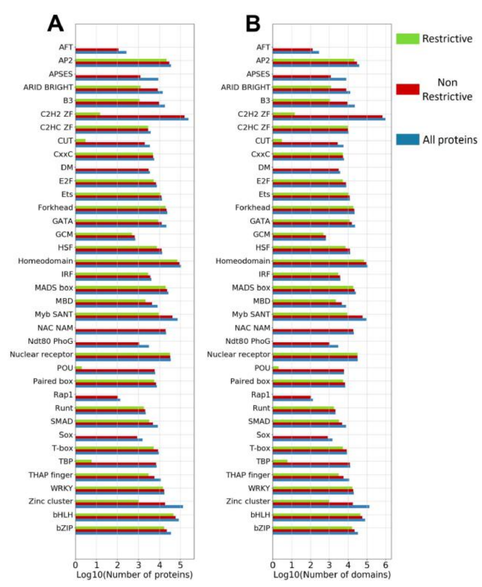

Figure 6: Total number of proteins (A) and domains (B) for which a PWM can be predicted using homology modelling in ModCRE (88.7%).